

1 Time penalties and Hänni’s collapse

Hänni’s working notes (Hänni, 2026)¹ compare axiom induction, which weighs axiom systems that prove the labelled data, with Solomonoff function induction, which weighs programs that label statements directly. He argues that the two are equivalent for consistent assigners, turning an assigner T into the schema “ $T(\text{quoted-}\phi) = \text{accept} \Rightarrow \phi$ ”, where “being an axiom of the right form is in fact decidable”. So unrestricted axiom induction collapses into function induction; this motivates his restriction to substitution schemas. The question adds a time penalty: “it should not take too long to generate axioms / to check whether a statement is an axiom”. The brief (H6) conjectured that such a penalty prices the collapse schema by the assigner’s running time.

Following model §6, we show that the equivalence holds within a constant (via Craig’s trick; via his own schema only in a two-sorted reading), that templates block his schema but not the collapse, that time penalties on checking or generating axioms do not block it, and that derivation length in written symbols prices it, while plain L1 does so only logarithmically as far as proved. Remark 1.20 states the result. Proofs are in Appendix A.

1.1 The two inductors

Fix a countable first-order language L with sentences Sent_L , a universal prefix machine U (program p has weight $2^{-|p|}$), and a decidable background $B \subseteq \text{Sent}_L$, possibly empty, that is part of every axiom hypothesis. Data are labelled sentences $D = ((\varphi_1, b_1), \dots, (\varphi_n, b_n))$, $b_i \in \{1, 0\}$ (true, false).

Definition 1.1 (Axiom and function induction). model §6.1

- **AI**: a hypothesis is a program p that halts on every input and decides a set $A_p \subseteq \text{Sent}_L$. It is *compatible* with D if $B \cup A_p$ is consistent and proves φ_i when $b_i = 1$ and $\neg\varphi_i$ when $b_i = 0$. $W_{\text{AI}}(D) := \sum_{p \text{ compatible}} 2^{-|p|}$.
- **FI_{cons}**: a hypothesis is a program f computing a partial map $\text{Sent}_L \rightarrow \{\text{acc}, \text{rej}\}$ with $B \cup \Gamma_f$ consistent, where $\Gamma_f := \{\varphi : f(\varphi) = \text{acc}\} \cup \{\neg\varphi : f(\varphi) = \text{rej}\}$; it is compatible if it gives each φ_i the label b_i ; $W_{\text{FI}}(D) := \sum_{f \text{ compatible}} 2^{-|f|}$. **FI_{all}** is the same over all assigners.
- **Semimeasure convention**: $P_X(b_1 \dots b_n \mid \varphi_1 \dots \varphi_n) := W_X(D_n) / W_X(\emptyset)$, and log loss is $-\log_2$ of it. Hypotheses that leave φ undecided drop out, so $W(D + (\varphi, 1)) + W(D + (\varphi, 0)) \leq W(D)$ (Hänni: “maybe we should speak of a semimeasure here instead of renormalizing”). $W_X(D)$ depends on D only as a set.

Proposition 1.2 (Why consistent assigners). [proved]^{model Prop 6.0} *Let R be a unary predicate, $w_1, w_2, \dots$ distinct closed terms, and φ_i be $R(w_i)$ or $\neg R(w_i)$ by independent fair coins, all labelled true (a consistent labelling). **FI_{all}**’s log loss is at most a constant (the program “always accept”); the expected log loss of **AI** and of **FI_{cons}** is at least n bits.*

Proof idea. No compatible **AI** or **FI_{cons}** hypothesis labels both ψ and $\neg\psi$ true, so the two predictive probabilities sum to at most 1; by convexity a step costs at least one bit in expectation (Appendix A.1). This confirms Hänni’s informal separation, and is why the fair comparison is with **FI_{cons}**. $\square$

¹File `research/prior/hanni-solomonoff-axiom-induction.md` of the accompanying repository; working notes, quoted verbatim.

Theorem 1.3 (Hänni’s equivalence, repaired). *[proved; Craig’s theorem known]^{model Thm 6.1} There is a constant c , depending only on U , the coding of syntax and the decider for B , such that $2^{-c}W_{\text{AI}}(D) \leq W_{\text{FI}}(D) \leq 2^c W_{\text{AI}}(D)$ for every finite labelled D . Hence, for every reveal order and every labelling, the log losses of AI and FI_{cons} in the semimeasure convention differ by at most $2c$ bits.*

Proof idea. $\text{AI} \rightarrow \text{FI}$ is Hänni’s argument: map p to “search for proofs of φ and of $\neg\varphi$ from $B \cup A_p$ ”. For $\text{FI} \rightarrow \text{AI}$ his schema is replaced by Craig’s trick (Craig, 1953): f becomes a decider of $A_f^C := \{\varphi^{\wedge(k+1)} : f(\varphi) = \text{acc in exactly } k \text{ steps}\} \cup \{(\neg\varphi)^{\wedge(k+1)} : f(\varphi) = \text{rej in exactly } k \text{ steps}\}$, $\psi^{\wedge m}$ the right-nested m -fold conjunction. A_f^C is decidable and has the consequences of Γ_f ; its decider is f plus a fixed wrapper (Appendix A.1). $\square$

Remark 1.4 (The convention, and “what is a sequence”). *[proved]^{model §6.1} Renormalising at each step, $P(b \mid \varphi, D) := W(D + (\varphi, b)) / \sum_{b'} W(D + (\varphi, b'))$, the argument gives only $2c$ bits *per step*. The constant equivalence needs the semimeasure convention, which answers Hänni’s “wtf is a sequence here?” (any reveal order, any labelling). His “same spec length as the axiom system plus a const” holds in both directions, and B needs no arithmetic; his route through the schema does, and works only with two sorts (§1.2).*

1.2 Hänni’s schema

Let $\text{Accept}_f(x)$ be a Σ_1 formula of arithmetic for “ f halts on input x with output acc” (via Kleene’s T predicate), and $\text{Reject}_f(x)$ likewise. Hänni’s schema is $A_f := \{\text{Accept}_f(\ulcorner\varphi\urcorner) \rightarrow \varphi : \varphi \in \text{Sent}_L\} \cup \{\text{Reject}_f(\ulcorner\varphi\urcorner) \rightarrow \neg\varphi : \varphi \in \text{Sent}_L\}$, and his consistency argument is that “a model of the phis alone would be a model of the added sentences also”.

Proposition 1.5 (Two sorts: his argument works). *[proved; Σ_1 -completeness of Q known]^{model Prop 6.2} Let L -sentences live in one sort and arithmetic in another, with $B := \text{Q}$ on the second. If Γ_f has a model, then $B \cup A_f$ is consistent and $\text{Cn}(B \cup A_f) \cap \text{Sent}_L = \text{Cn}(\Gamma_f)$.*

Proposition 1.6 (One sort over PA). *[proved, assuming $\text{Con}(\text{PA})$ and the standard formalisation of computations; Hänni’s consistency argument refuted in this reading]^{model Prop 6.3} Let $L = L_A$, $B = \text{PA}$, and let f accept $\ulcorner\neg\text{Con}(\text{PA})\urcorner$, search on input $\ulcorner\text{Con}(\text{PA})\urcorner$ for a PA-proof of $\perp$ and accept if it finds one, and loop on every other input. Then $\Gamma_f = \{\neg\text{Con}(\text{PA})\}$ is consistent with PA, but $\text{PA} \cup A_f$ is inconsistent.*

Proof idea. $\text{PA} \cup A_f$ proves $\neg\text{Con}(\text{PA})$ from the true Σ_1 sentence $\text{Accept}_f(\ulcorner\neg\text{Con}(\text{PA})\urcorner)$, and $\text{Con}(\text{PA})$ from $\text{PA} \vdash \neg\text{Con}(\text{PA}) \rightarrow \text{Accept}_f(\ulcorner\text{Con}(\text{PA})\urcorner)$ (Appendix A.2). $\square$

His argument assumes that the antecedents are evaluated correctly; in a model of $\text{PA} + \neg\text{Con}(\text{PA})$, $\text{Accept}_f(\ulcorner\text{Con}(\text{PA})\urcorner)$ holds via a nonstandard proof of $\perp$. So his *conclusion*, the equivalence, stands (Theorem 1.3); his *argument* needs standard arithmetic (two sorts) or a sound f .

1.3 What templates do

Proposition 1.7 (No non-ground template in the collapse schema). *[proved]^{model Prop 6.4; pa Prop 4.3} Write Gödel numbers as unary numerals S^k0 (or in any numeral system in which some term symbol, here $+$, never occurs), and let $C_f := \{\text{Accept}_f(\ulcorner\varphi\urcorner) \rightarrow \varphi : \varphi \in \text{Sent}_L\}$, single- or two-sorted. Every DT° template τ with $\text{inst}(\tau) \subseteq C_f$ is ground. So C_f is not $\bigcup \text{inst}(T)$ for any finite*

set T of DT° templates, and such a set with instances in C_f covers finitely many of its members. The same holds for the reflection schema $\text{Refl}_T = \{\text{Prv}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi\}$ (??).

Proof idea. A metavariable at the root or in the antecedent takes a constant body that breaks the shape $\text{Accept}_f(\text{numeral})$ (AS Lemma 2.4). Otherwise the antecedent is fixed, which fixes φ , so τ has one instance, while a template with a metavariable has infinitely many (AS Thm 2.5(a); Appendix A.3). $\square$

So Hänni’s restriction (“maybe we just say we have to have second-order substitution schemas of a specific kind”) blocks his construction: inside DT° it costs one ground template per sentence, which is memorisation. But one ground sentence can carry the collapse, at least in arithmetic.

Proposition 1.8 (One reflection sentence carries the collapse). *[proved, given partial truth predicates (known)]^{model Prop 6.5} For $n \geq 1$ let $\text{Tr}_n(x)$ be a Σ_n partial truth predicate, $\text{PA} \vdash \varphi \leftrightarrow \text{Tr}_n(\ulcorner \varphi \urcorner)$ for every Σ_n sentence φ (known; Hájek and Pudlák, 1993, Ch. I; Kaye, 1991, Ch. 9; chapter numbers not checked), and $\text{Sent}_{\Sigma_n}(x)$ a Δ_1 formula for “ x codes a Σ_n sentence”. The single sentence (ground DT° template)*

$$\rho_{f,n} := \forall x[(\text{Sent}_{\Sigma_n}(x) \wedge \text{Accept}_f(x)) \rightarrow \text{Tr}_n(x)] \wedge \forall x[(\text{Sent}_{\Sigma_n}(x) \wedge \text{Reject}_f(x)) \rightarrow \neg \text{Tr}_n(x)]$$

has $\ell(\rho_{f,n}) \leq a|f| + b_n$ for constants a, b_n (binary numeral for f ’s index), and: (a) $\text{PA} + \rho_{f,n}$ proves every Σ_n sentence that f accepts and the negation of every one it rejects; (b) if f is Σ_n -sound (it accepts only true and rejects only false Σ_n sentences), $\text{PA} + \rho_{f,n}$ is true in $\mathbb{N}$.

So PA ’s templates plus one ground sentence reproduce any Σ_n -sound assigner on Σ_n sentences, at a cost linear in $|f|$ (not $|f| + c$); for merely consistent assigners $\rho_{f,n}$ can be inconsistent with PA (as in Proposition 1.6). It is also why the toy tower of ?? adds one ground sentence per step.

1.4 Time penalties

Proposition 1.9 (Membership is cheap in both collapse sets). *[(a), (b) proved; the brief’s H6 sentence refuted]^{model Prop 6.6} (a) With binary numerals, membership of χ in A_f is decidable in time polynomial in $|\chi|$ without running f : parse χ as $\text{Accept}_f(t) \rightarrow \varphi$ or $\text{Reject}_f(t) \rightarrow \neg \varphi$, compute $\ulcorner \varphi \urcorner$, compare with t . With unary numerals likewise, since then $|\chi| \geq \ulcorner \varphi \urcorner$. (b) Membership in A_f^C is decidable in time polynomial in $|\chi|$: at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$, each running f for $m - 1 < |\chi|$ steps.*

So a penalty in the style of Levin’s Kt or the speed prior on the time to *check* axiomhood puts these sets in the same polynomial class as DT° templates (??(iii)). The brief’s sentence “A schema ‘ $T(\ulcorner \varphi \urcorner) = \text{accept} \rightarrow \varphi$ ’ requires running T to check membership, so a Kt- or speed-prior-style penalty prices it by T ’s running time” is *refuted* by (a). Penalising the time to *generate* axioms fails too: $\varphi^{\wedge(k+1)} \in A_f^C$ comes from k steps of f and is longer than k .

The computation of f moves into the proofs (with A_f a derivation must prove $\text{Accept}_f(\ulcorner \varphi \urcorner)$) or into the length of the axioms (A_f^C). Both are counted by the symbol size of derivations, and no theory escapes this. Let $L \supseteq \{0, S, +, \cdot, R\}$, R unary. For $X \subseteq \{0, 1\}^*$ put $\varphi_w := R(\underline{w})$, $\underline{w}$ a binary numeral term, and let f_X accept φ_w iff $w \in X$ and reject it otherwise; Γ_{f_X} is consistent (interpret R as X).

Theorem 1.10 (Derivation size bounds the assigner’s nondeterministic time). *[proved]^{model Thm 6.7} Let T be any axiom set with membership decidable in time $O(n^e)$ and $T \cup \Gamma_{f_X}$ consistent. If every*

φ_w , $w \in X$, has a K -derivation from T of symbol size at most $\ell(|w|)$, with $\ell(m) \geq m$ time-constructible, then $X \in \text{NTIME}(\ell(m)^{c_e})$, where c_e depends only on e and on K 's proof-checking overhead.

Proof idea. Guess a derivation of size at most $\ell(|w|)$ and check it. If $w \notin X$, then $\neg\varphi_w \in \Gamma_{f_X}$, so consistency rules out a derivation (Appendix A.4). $\square$

Corollary 1.11 (Hard assigners force long derivations, for every theory). *[proved, given the non-deterministic time hierarchy theorem (known); single X for all e : proof sketch]^{model Cor 6.8 (referee m1)} For finite DT° theories membership takes time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ and c_1 is one constant for all of them. (a) Let $s(m) \geq m$ be time-constructible with $s(m+1)^{c_1} = o(s(m)^{c_1+1})$ (every polynomial). There is a decidable X such that every finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent has, for infinitely many m , some $w \in X$ of length m with no K -derivation of φ_w from T of symbol size $\leq s(m)$. (b) For each e the same holds, with X depending on e , for every axiom set with membership in time $O(n^e)$; a single X for all e exists under a growth condition (proof sketch, Appendix A.4).*

The hierarchy theorem is cited from Cook (1972); Seiferas et al. (1978); Žák (1983); its exact form was not checked. X has a fixed program, so f_X is cheap for function induction and expensive in derivation size for every template theory. (The pre-referee version let c depend on T ; referee m1.)

Remark 1.12 (Upper bounds). *[proof sketch]^{model §6.4} The collapse constructions pay at most a polynomial of the assigner's deterministic running time k in derivation size: cite $\varphi^{\wedge(k+1)}$ and eliminate $\wedge$; or prove the true Σ_1 sentence $\text{Accept}_f(\ulcorner\varphi\urcorner)$ in $\mathcal{Q}$, in size polynomial in the computation (recalled from Pudlák, 1998, not checked), then apply MP (two sorts), plus the Tarski biconditional for $\rho_{f,n}$.*

1.5 What the derivation likelihoods charge

Theorem 1.10 bounds the *symbol* size of derivations. The code length $-\ln \Pr(\pi)$ of an L1 tree counts its grammar choices instead (??), and the two can differ exponentially.

Remark 1.13 (Refuted: plain L1 charges symbol size). *[refuted; counterexample proved and computed]^{model Prop 6.9 (referee M2)} Claim (pre-referee, as a sketch): under L1 with subcritical parameters, $-\ln P_T(\varphi) \geq \kappa \ell_{\min}(\varphi) - \ln(C/Z_T)$ for constants $\kappa, C > 0$, with $\ell_{\min}(\varphi)$ the least symbol size of a derivation of φ from T . Counterexample. From the logical axiom $\forall x (x = x)$, repeat k times “ $\forall E$ with $t = p+p$, then Gen on p ”, and close with one $\forall E$. The conclusion φ_k has size exactly $2^{k+3} - 1$; the tree has $7+8k$ grammar nodes and, in the referee's grammar, $-\ln \Pr(\text{tree}) \leq 12.8+12.2k$ nats. Every derivation of φ_k contains φ_k , while $-\ln P_T(\varphi_k) \leq -\ln \Pr(\text{tree})$ (as $Z_T \leq 1$) grows linearly in k . Checked by two independent scripts (Table 1); a one-node A4 example is in Appendix A.5.*

$\forall E$ and Gen substitute into every occurrence of a variable at the price of one choice. So Theorem 1.10 and Corollary 1.11 transfer to L1^σ and to Hänni's graded score, not to plain L1. Let $\nu(\pi)$ count the grammar nodes of a tree π (derivation nodes, $\mathcal{Q}$ nodes of bodies and terms, nodes of the geometric chains that choose parameters), and $\nu_{\min}(s)$ be the least ν of a valid tree concluding s .

Lemma 1.14 (Symbol size against grammar choices). *[proved; computed]^{model Lemma 6.10} Let π be a valid L1 tree from a finite T with ν grammar nodes, $a_T \geq 2$ a bound on the sizes of the templates of T and of K 's logical templates, and $c_T := 2a_T^2$. Every conclusion in π has symbol size at most $(c_T + 1)\nu^2 e^{\nu/e}$.*

Proof idea. $\forall E$ multiplies the size by at most $|t|$; the substituted terms are disjoint sets of grammar nodes, so by AM–GM their sizes multiply to at most $e^{\nu/e}$ (Appendix A.5, with a random-chain check). $\square$

Proposition 1.15 (L1 charges its own code length exponentially). *[proved; computed]^{model Prop 6.11} Assume L1 with $m < 1$, $\mathcal{Q}$ uniformly subcritical (??), T finite and a geometric parameter law. There are $\zeta, C > 0$, depending on the grammar and on T , with $P_T(\varphi) \leq C e^{-\zeta \nu_{\min}(\varphi)} / Z_T$ for every φ .*

Proof idea. The whole tree satisfies the uniform condition, so $\mathbb{E}[e^{\zeta \nu}] \leq C$ (??(d)); then Markov’s inequality. $\square$

Theorem 1.16 (Code length bounds the assigner’s time, logarithmically). *[proved]^{model Thm 6.12} In the setting of Theorem 1.10, let T be a finite DT° theory with $T \cup \Gamma_{f_X}$ consistent, and let every φ_w , $w \in X$, have a valid L1 tree from T with at most $\ell(|w|)$ grammar nodes, $\ell(m) \geq m$ time-constructible. Then $X \in \text{NTIME}(2^{(d+1)\ell(m)})$, where d , the polynomial degree of the syntactic operations used in checking, depends neither on T nor on ℓ .*

Proof idea. Guess the tree and write every conclusion out; by Lemma 1.14 it has size at most $(c_T + 1)\ell^2 e^{\ell/e}$, and $e^{d/e} < 2^d$ (Appendix A.5). $\square$

Corollary 1.17 (Hard assigners make plain L1 pay, logarithmically). *[proved, given the hierarchy theorem (known)]^{model Cor 6.13} Let $s(m) \geq m$ be time-constructible with $s(m)^2 - (d+1)s(m+1) \rightarrow \infty$ (every polynomial; 2^m). There is a decidable X such that for every finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent, for infinitely many m some $w \in X$ of length m has $\nu_{\min}(\varphi_w) > s(m)$; under the hypotheses of Proposition 1.15, $-\ln P_T(\varphi_w) \geq \zeta s(m) - \ln(C/Z_T)$ on those data.*

So if the assigner’s language needs nondeterministic time $t(m)$, plain L1 is guaranteed to charge only about $\ln t(m)$ nats per hard datum; symbol-size penalties charge a polynomial root of $t(m)$:

Proposition 1.18 (Symbol-size penalties charge polynomially). *[(a) proved; (b), (c) proved, given the hierarchy theorem (via Corollary 1.11)]^{model Prop 6.14} (a) Under L1^σ (??), $P_T^\sigma(\varphi) \leq e^{-\kappa \ell_{\min}^\sigma(\varphi)} Z_T / Z_T^\sigma$, with $\ell_{\min}^\sigma(\varphi)$ the least total symbol size of a valid L1 tree concluding φ . (b) A valid L1 tree of total symbol size ℓ yields a K-derivation of symbol size at most 4ℓ . So with X from Corollary 1.11(a), for every finite DT° theory T consistent with Γ_{f_X} , infinitely often $-\ln P_T^\sigma(\varphi_w) \geq \kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$. (c) Hänni’s graded score with $g(\ell) = 2^{-\kappa \ell}$ on explicit K-derivations, in a code with at least one bit per written symbol (??), charges at least $\kappa s(m)$ bits on the same data, and at least $\kappa s(m) + \log_2 Z_T$ after normalisation.*

Conjecture 1.19 (Plain L1 charges polynomially too). *[conjecture]^{model Conj 6.15} Theorem 1.16 holds with $\text{NTIME}(\ell(m)^c)$, for a constant c , in place of $\text{NTIME}(2^{(d+1)\ell(m)})$.*

The route is to check trees on shared representations (formula DAGs, compressed terms); the missing step is de Bruijn indices and Gen under sharing (Appendix A.5).

1.6 The precise statement, and links

Remark 1.20 (What the time penalty does). *[proved; upper bound in (3): proof sketch]^{model §6.5} (1) Penalties on the time to check or generate axioms (Kt, speed prior) do not block the collapse: with Craig’s sets they change Hänni’s equivalence by an additive constant at most (Theorem 1.3*

and Proposition 1.9). (2) Templates block his schema but not the collapse in arithmetic (Propositions 1.7 and 1.8). (3) Derivation size in written symbols prices the assigner per datum, between its nondeterministic time (a lower bound for every theory, Theorem 1.10 and Corollary 1.11) and its deterministic time (an upper bound for the collapse constructions, Remark 1.12): $L1^\sigma$ charges at least $\kappa s(m)/4$ nats and the graded score at least $\kappa s(m)$ bits on infinitely many hard data (Proposition 1.18). (4) Plain $L1$ is proved to charge at least $\zeta s(m)$ nats when the assigner’s language lies outside $\text{NTIME}(2^{(d+1)s})$, i.e. logarithmically in its time (Corollary 1.17); a polynomial charge is Conjecture 1.19. By contrast, function induction has $W_{\text{FI}}(D_n) \geq 2^{-|f_X|}$ when $B \cup \Gamma_{f_X}$ is consistent, so its total log loss on f_X ’s labels is at most $|f_X|$ bits.

So the question’s derivation-length prior should measure length in written symbols ($L1^\sigma$, or the graded score normalised as in ??), unless Conjecture 1.19 is proved. Whether axiom induction with a symbol-size penalty is equivalent, up to polynomial factors, to function induction over consistent assigners with short certificates is open (model §10, problem 13).

Remark 1.21 (Links to the other sections). [(i) proved; (ii) and the argument of (iv): proof sketch; (iii), the numbers of (iv): computed]^{universal Prop U2t, §8; experiments E7; pa §3.8, F9} (i) No penalty monotone in derivation length or time reverses ?? : in $C_{\min}$ a chain weight non-increasing in the number of eliminations keeps $P_{H_V}^f(s) \leq P_{H_{\text{sch}}}^f(s)$ on every datum $s \neq \forall x\varphi$ (??, U2t). This counts steps, not symbols. (ii) In the equational fragment $\{Q4, Q5\}$, a Kt-style penalty on generating $0 + S^k 0 = S^k 0$ charges $O(\ln k)$, less than the derivation charge of ?? (at least $k + 1$ axiom leaves). (iii) The experiments’ membership-time factor $(1 + |T|)^{-\tau}$ is a few bits by construction ($\leq \log_2(78/2) \approx 5.3$ bits against T^* at time exponent $\tau = 1$); on the E2 data it changes nothing visible from $n = 64$ and slightly favours small unsound lumps at $n \leq 32$ (unsound MAP at $n = 8$ in 9 instead of 7 of 25 seeds; ??). (iv) A time-bounded evaluation of the likelihood sums over the derivations found within the bound, so it lower-bounds $\mu_T(s)$ and favours adopting theorems with long proofs as axioms; a two-part code is the extreme case. Under $L1^{\text{sch}}$ memorising already beats deriving at the first occurrence for 11-line proofs (??, ??). (ii) and (iv) agree: a derivation likelihood favours the theory that makes the data cheap to state, not the logically minimal one. The PA comparison proposed by pa’s referee ($I\Sigma_n$ as Q plus one partial-truth sentence, against the schema, under a derivation-time penalty) was not quantified.

Remark 1.22 (A steeper simplicity penalty). [computed]^{experiments E7} On the E2 data (25 seeds), prior steepness $\lambda = 2$ makes the MAP at $n = 8$ unsound in 22 seeds ($\lambda = 1$: 7) but removes the false spare slot faster ($2 \cdot 10^{-4}$ against 0.004 unsound mass at $n = 512$); $\lambda = \frac{1}{2}$ (proper only on finite pools) does the reverse; neither removes both failure modes (??).

Conjecture 1.23 (A time-bounded analogue). [conjecture; not written out]^{model §9.3} *Hänni’s note on time-bounded Solomonoff induction² gives, for each polynomial p , a predictor whose log loss exceeds that of any “prediction algorithm P with runtime $O(p(n))$ ” by “at most a constant c_P ”. The analogue here is the derivation-bounded $L2$ inducer with a growing depth budget; a constant-regret bound against theories whose data have derivations within the budget plausibly follows by his argument.*

A Time penalties: proofs and details

Scripts cited here are in the folders **checks/** (the model track’s) and **referee_code/** (its referee’s) of **research/tracks/model/**; each is seeded and writes its output next to itself. Sizes are symbol

²File **research/prior/hanni-polytime-solomonoff.md**; working notes, in which he says he has not “checked all the technical claims completely carefully”.

counts in closure-normal form with de Bruijn indices: a quantifier, a connective, a function or relation symbol, a constant, an index and a parameter each count one.

A.1 The two inductors

Proof of Proposition 1.2. FI_{all} : the program “always accept” has some length c and labels every φ_i true, so $W_{\text{FI}_{\text{all}}}(D_n) \geq 2^{-c}$; with $W_{\text{FI}_{\text{all}}}(\emptyset) \leq 1$ (Kraft, for a prefix machine) the log loss $-\log_2 W(D_n) + \log_2 W(\emptyset)$ is at most c . AI and FI_{cons} : let D be the data so far and $\psi := R(w_i)$ for the next index. A hypothesis compatible with $D + (\psi, 1)$ and with $D + (\neg\psi, 1)$ would prove, or accept, both ψ and $\neg\psi$, against consistency. So the two sets of compatible hypotheses are disjoint subsets of those compatible with D , and the predictive probabilities $a := W(D + (\psi, 1))/W(D)$ and $a' := W(D + (\neg\psi, 1))/W(D)$ satisfy $a + a' \leq 1$. The coin shows ψ or $\neg\psi$ with probability $\frac{1}{2}$ each, independently of D , so the expected step loss is $\frac{1}{2}(-\log_2 a - \log_2 a') \geq -\log_2 \frac{a+a'}{2} \geq 1$ bit, by convexity of $-\log_2$. Summing over the n steps gives at least n bits. $\square$

Proof of Theorem 1.3. $W_{\text{FI}} \geq 2^{-c'} W_{\text{AI}}$. For a decider p let f_p be the program: on input φ , dovetail searches for proofs of φ and of $\neg\varphi$ from $B \cup A_p$ (both sets are decidable, so the theorems are recursively enumerable), and output acc or rej for the first proof found. Its code is a fixed wrapper followed by p , so $|f_p| \leq |p| + c'$ and $p \mapsto f_p$ is injective. Let p be compatible with D . Then $B \cup A_p$ is consistent, so at most one of φ , $\neg\varphi$ is provable, and f_p labels each φ_i with b_i . Also $\Gamma_{f_p} \subseteq \text{Th}(B \cup A_p)$, so $B \cup \Gamma_{f_p}$ is consistent and f_p is a compatible FI_{cons} hypothesis. Summing $2^{-|f_p|} \geq 2^{-c'} 2^{-|p|}$ over compatible p gives the claim.

$W_{\text{AI}} \geq 2^{-c''} W_{\text{FI}}$ (*Craig’s trick*). For an FI_{cons} hypothesis f define A_f^C as in Theorem 1.3. Its decider p_f : given χ , try each of the at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$; for each, run f for $m - 1$ steps on ψ and check whether it accepts in exactly $m - 1$ steps, and, if $\psi = \neg\varphi$, run f on φ and check whether it rejects in exactly $m - 1$ steps. Every run is bounded, so p_f halts on every input, even when f is partial. Each member $\varphi^{\wedge(k+1)}$ is logically equivalent to $\varphi \in \Gamma_f$ (likewise for $\neg\varphi$), and each member of Γ_f has such a power in A_f^C , with k the halting time. So $\text{Cn}(B \cup A_f^C) = \text{Cn}(B \cup \Gamma_f)$, which is consistent, and $B \cup A_f^C$ decides every φ_i as f does. Hence p_f is compatible whenever f is. Its code is a fixed wrapper followed by f : $|p_f| \leq |f| + c''$, and $f \mapsto p_f$ is injective.

Losses. Put $c := \max(c', c'')$. The semimeasure log loss of an inducer on D_n is $-\log_2 W(D_n) + \log_2 W(\emptyset)$; each of the two terms differs by at most c between AI and FI_{cons} , and both depend on D_n only as a set. If one weight is 0, so is the other, and both losses are infinite. $\square$

Per-step renormalisation (Remark 1.4). With $P_X(b \mid \varphi, D) := W_X(D + (\varphi, b)) / (W_X(D + (\varphi, 0)) + W_X(D + (\varphi, 1)))$, the numerator and the denominator each change by a factor in $[2^{-c}, 2^c]$ between the inducers, so each step’s ratio lies in $[2^{-2c}, 2^{2c}]$; over n steps the bound is $2cn$ bits, and the argument gives nothing better.

A.2 Hänni’s schema

Proof of Proposition 1.5. Consistency. Take the structure whose arithmetic sort is the standard $\mathbb{N}$ and whose L -sort is a model M of Γ_f . In $\mathbb{N}$, $\text{Accept}_f(\ulcorner \varphi \urcorner)$ is true iff f accepts φ , and then $\varphi \in \Gamma_f$ holds in M ; likewise for rejections. So every member of A_f holds, and so does $\mathbf{Q}$. $\supseteq$. If $f(\varphi) = \text{acc}$, then $\text{Accept}_f(\ulcorner \varphi \urcorner)$ is a true Σ_1 sentence, hence provable in $\mathbf{Q}$ (Σ_1 -completeness; known, e.g. Boolos et al., 2007, chapter on $\mathbf{Q}$, not checked); MP with its member of A_f gives φ . Rejections are similar. So $\Gamma_f \subseteq \text{Cn}(B \cup A_f)$. $\subseteq$. An L -sentence provable from $B \cup A_f$ holds in

every structure $\mathbb{N} \oplus M$ with $M \models \Gamma_f$; its truth there depends only on M . So it holds in every model of Γ_f , and by completeness it lies in $\text{Cn}(\Gamma_f)$. $\square$

Proof of Proposition 1.6. Γ_f . Since $\text{Con}(\text{PA})$ holds, the search on $\ulcorner \text{Con}(\text{PA}) \urcorner$ never ends, so $\Gamma_f = \{\neg \text{Con}(\text{PA})\}$. $\text{PA} + \neg \text{Con}(\text{PA})$ is consistent by Gödel's second incompleteness theorem. $\neg \text{Con}(\text{PA})$. $\text{Accept}_f(\ulcorner \neg \text{Con}(\text{PA}) \urcorner)$ is a true Σ_1 sentence, so PA proves it; with its member of A_f , $\text{PA} \cup A_f \vdash \neg \text{Con}(\text{PA})$. $\text{Con}(\text{PA})$. $\text{PA} \vdash \neg \text{Con}(\text{PA}) \rightarrow \text{Accept}_f(\ulcorner \text{Con}(\text{PA}) \urcorner)$: if a proof of $\perp$ exists there is a least one, and PA proves that the search halts on it with output acc . This formalisation step is standard and already provable in IS_1 . With the member $\text{Accept}_f(\ulcorner \text{Con}(\text{PA}) \urcorner) \rightarrow \text{Con}(\text{PA})$ of A_f , $\text{PA} \cup A_f \vdash \neg \text{Con}(\text{PA}) \rightarrow \text{Con}(\text{PA})$, so $\text{PA} \cup A_f \vdash \text{Con}(\text{PA})$. Both together are a contradiction. $\square$

A.3 Templates

Proof of Proposition 1.7. Let $\tau \in \text{DT}^\circ$ with $\text{inst}(\tau) \subseteq C_f$, and suppose τ has a metavariable M with an occurrence at position q . Constant bodies exist for every type with two different root symbols: $\lambda \bar{z}.0$ and $\lambda \bar{z}.(0 + 0)$ for terms, $\lambda \bar{z}.(0 = 0)$ and $\lambda \bar{z}.\neg(0 = 0)$ for formulas. By the preservation lemma (AS Lemma 2.4), for a constant body the subtree of $\tau\theta$ at q is that constant, whatever the occurrence's arguments, and the rigid symbols of τ above q are the same in every instance. Every member of C_f has the form $\text{Accept}_f(N) \rightarrow \varphi$, where N is the numeral $\ulcorner \varphi \urcorner$ substituted for the free variable of the fixed formula $\text{Accept}_f(x)$. Since $\text{inst}(\tau) \neq \emptyset$, the path from the root to q is a path of such a member.

Case 1: q is the root or lies in the antecedent. (i) If q lies inside an occurrence of the numeral N (its root included), every member of C_f carries S or 0 at q . Put $\theta(M) := \lambda \bar{z}.(0 + 0)$ and any bodies elsewhere. Then $\tau\theta$ has $+$ at q . If $\tau\theta$ were in C_f , q would again lie inside a numeral, since the path to q is the same; so $\tau\theta \notin C_f$. (ii) Otherwise q is the root or a position of the fixed skeleton of $\text{Accept}_f(x)$. All members of C_f carry the same symbol there ($\rightarrow$ at the root). Choose the constant body of the right type whose root symbol differs from it; then $\tau\theta \notin C_f$. Both contradict $\text{inst}(\tau) \subseteq C_f$.

Case 2: no metavariable occurs at the root or in the antecedent. Then the antecedent of every instance is one fixed sentence $\text{Accept}_f(N_0)$, and the only member of C_f with that antecedent is $\text{Accept}_f(\ulcorner \varphi_0 \urcorner) \rightarrow \varphi_0$ for the single φ_0 with $\ulcorner \varphi_0 \urcorner = N_0$. So τ has at most one instance. But a template with a metavariable has infinitely many instances: each type has infinitely many bodies, and $\theta \mapsto \tau\theta$ is injective (AS Thm 2.5(a)). Contradiction.

So τ is ground. A ground template has one instance; C_f is infinite, so no finite set of DT° templates has instance union C_f , and such a set with instances in C_f covers at most as many members as it has templates.

Reflection schema (pa Prop 4.3). With $\text{Prv}_T(x)$ in place of $\text{Accept}_f(x)$ the members are $\text{Prv}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$, with unary numerals, and the proof goes through word for word. $\square$

Implicit in the notes. Case 1(i) uses that “inside the numeral” is determined by the path to q , which is rigid in τ ; we have made this explicit. The notes, and the statement, treat the Accept_f part C_f ; the Reject_f part is symmetric.

Proof of Proposition 1.8. (a) Let $f(\varphi) = \text{acc}$ with $\varphi \in \Sigma_n$. Then $\text{PA} \vdash \text{Accept}_f(\ulcorner \varphi \urcorner)$ (a true Σ_1 sentence) and $\text{PA} \vdash \text{Sent}_{\Sigma_n}(\ulcorner \varphi \urcorner)$ (a true Δ_1 fact). So $\text{PA} + \rho_{f,n} \vdash \text{Tr}_n(\ulcorner \varphi \urcorner)$, and $\text{PA} \vdash \text{Tr}_n(\ulcorner \varphi \urcorner) \rightarrow \varphi$. For rejections, $\text{PA} + \rho_{f,n} \vdash \neg \text{Tr}_n(\ulcorner \varphi \urcorner)$ and $\text{PA} \vdash \varphi \rightarrow \text{Tr}_n(\ulcorner \varphi \urcorner)$, so $\neg \varphi$ follows. (b) In $\mathbb{N}$, Accept_f and Reject_f hold exactly of f 's actual decisions, and Tr_n is Σ_n -truth, so both conjuncts of $\rho_{f,n}$ are true when f is Σ_n -sound. *Length.* f enters $\rho_{f,n}$ only through its index,

written as a binary numeral of length $O(|f|)$ inside Accept_f and Reject_f ; the rest of the sentence depends on n only. $\square$

A.4 Time penalties

Proof of Proposition 1.9. (a) Parsing χ as an implication with antecedent $\text{Accept}_f(t)$ or $\text{Reject}_f(t)$ is a comparison with a fixed formula, linear in $|\chi|$. Computing the Gödel number of the consequent (or of the formula under its negation) is polynomial in $|\chi|$, and so is writing it as a binary numeral and comparing it with t . With unary numerals, t itself has length $\lceil \varphi^\top \rceil \leq |\chi|$, so writing out the numeral is again polynomial in $|\chi|$. No step runs f . (b) $\chi = \psi^{\wedge m}$ forces $m \leq |\chi|$, and there are at most $|\chi|$ candidate decompositions. For each, f is simulated for $m - 1 < |\chi|$ steps on ψ (and on φ if $\psi = \neg\varphi$), in time polynomial in $|\chi|$ for fixed f . $\square$

Proof of Theorem 1.10. Algorithm. On input w of length m , compute φ_w and $\ell(m)$ (time-constructible), guess a K -derivation of symbol size at most $\ell(m)$, and check it: each line is an instance of a logical axiom (A1–A3, A5 and the equality axioms by matching; A4 by first-order matching, which is polynomial), a member of T (time $O(\ell^e)$), or follows by MP or Gen. Accept iff the derivation is valid and ends in φ_w . All checks are polynomial in $\ell(m)$, of a degree that depends only on e and on the degree d_K of K 's line checks; so the machine runs in time $O(\ell(m)^{c_e})$. *Completeness.* If $w \in X$, such a derivation exists by hypothesis. *Soundness.* If $w \notin X$, then $\neg\varphi_w \in \Gamma_{f_X}$, so $T \vdash \varphi_w$ would make $T \cup \Gamma_{f_X}$ inconsistent. No derivation exists, and the machine rejects. $\square$

Proof of Corollary 1.11. Uniformity. The constant c_e of Theorem 1.10 depends on T only through e . Membership in a finite DT° theory is matching against each template, in time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ for all of them, with a T -dependent constant factor that does not change the class. *The hierarchy theorem* (known; exact form not checked against Cook, 1972; Seiferas et al., 1978; Žák, 1983): if t_2 is time-constructible and $t_1(m+1) = o(t_2(m))$, then $\text{NTIME}(t_2) \setminus \text{NTIME}(t_1) \neq \emptyset$. (a) Apply it with $t_1 := s^{c_1}$ and $t_2 := s^{c_1+1}$ to get a decidable $X \in \text{NTIME}(s^{c_1+1}) \setminus \text{NTIME}(s^{c_1})$. Suppose some finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent had, for all but finitely many m , a derivation of size at most $s(m)$ of every φ_w with $w \in X$, $|w| = m$. Put the finitely many words of the exceptional lengths in a table. The machine of Theorem 1.10 with $\ell = s$, plus the table, decides X in $\text{NTIME}(s^{c_1})$: a contradiction. (b) The same with c_e in place of c_1 , for each e . *A single X for all e* (proof sketch). Take $X \in \text{NTIME}(s^{\log s}) \setminus \text{NTIME}(s^{(\log s)/2})$, when these bounds are time-constructible and satisfy the gap condition. Every s^{c_e} is eventually below $s^{(\log s)/2}$, so the argument of (a) applies to every e . The time-constructibility and gap conditions for these bounds are not checked. $\square$

Upper bounds (Remark 1.12); proof sketch. From A_f^C , a derivation of φ cites $\varphi^{\wedge(k+1)}$, $k = \text{time}_f(\varphi)$, and applies $\wedge$ -elimination, in size polynomial in $k|\varphi|$. From A_f in two sorts, a derivation is a Q-proof of the true Σ_1 sentence $\text{Accept}_f(\lceil \varphi^\top \rceil)$ followed by MP; the size of such a Q-proof is polynomial in the length of the computation (recalled from Pudlák, 1998, not checked). From $\rho_{f,n}$, the same plus a PA-proof of the Tarski biconditional $\text{Tr}_n(\lceil \varphi^\top \rceil) \rightarrow \varphi$, whose size is not bounded in the notes.

A.5 What the derivation likelihoods charge

The counterexample to the pre-referee claim (Remark 1.13). Let $\varphi^{(0)} := \forall x(x = x)$, the logical axiom of reflexivity (size 4). Round $j = 1, \dots, k$: apply $\forall E$ with the term $p + p$ (p the

first parameter), then Gen on p ; this doubles the number of occurrences of the bound variable and keeps one $\forall$ at the root. A final $\forall E$ with $p+p$ gives φ_k , an equation whose two sides are balanced sums of 2^{k+1} copies of p ; its size is $2(2^{k+2} - 1) + 1 = 2^{k+3} - 1$. The tree has one citation node; each $\forall E$ contributes its own node and the 5 grammar nodes of $p+p$ (the $+$ node and two parameters, each with a one-node index chain), each Gen its node and a one-node chain: $\nu = 1 + 6 + 8k = 7 + 8k$. In the referee's grammar ($\alpha_{ax} = \alpha_{lg} = \alpha_{MP} = 0.2$, $\alpha_{Gen} = \alpha_{\forall E} = 0.1$, so $m = 0.6$; seven logical schemas cited uniformly; terms 0: .4, S : .2, $+$: .1, $\cdot$: .1, parameter: .2; parameter index geometric with ratio $\frac{1}{2}$) a round costs $-\ln(0.1 \cdot 0.1 \cdot (0.2 \cdot 0.5)^2) - \ln(0.1 \cdot 0.5) = 12.206$ nats and the citation plus final $\forall E$ 12.77 nats. Every derivation of φ_k contains φ_k , so $\ell_{\min}(\varphi_k) \geq 2^{k+3} - 1$, and $-\ln P_T(\varphi_k) \leq -\ln \Pr(\text{tree})$ because $Z_T \leq 1$. For every $\kappa > 0$ and C the claimed inequality fails for large k ; Table 1 gives the numbers.

One-node example. For $B_m(x) := (x + \dots + x = 0)$ with m summands and $t := S^m 0$, the A4 instance $\forall x B_m(x) \rightarrow B_m(t)$ has size quadratic in m but probability $e^{-O(m)}$: the referee's script gives sizes 145, 1765, 6725 symbols at costs at most 99.4, 369.0, 728.5 nats for $m = 10, 40, 80$ (r2_l1_size.out).

k	$\nu = 7 + 8k$	$ \varphi_k = 2^{k+3} - 1$	$-\ln \Pr(\text{tree})$ (nats)	$\ln \varphi_k / \nu$
0	7	7	12.77	0.278
1	15	15	24.97	0.181
3	31	63	49.38	0.134
10	87	8191	134.83	0.104
14	119	131071	183.65	0.099
30	247	$8.6 \cdot 10^9$	378.95	0.093

Table 1: The counterexample to the pre-referee claim of Remark 1.13. Sizes were built and checked by the referee's script for $k \leq 14$ (r2_l1_size.out; formula beyond) and independently by c12_l1_size.out, Part A, for $k \leq 12$, which also counts $\nu = 7 + 8k$. Costs are the referee's, for the grammar given in the text. $\ln |\varphi_k| / \nu \rightarrow \ln 2/8 \approx 0.087$, below the $1/e \approx 0.368$ that Lemma 1.14 allows.

Proof of Lemma 1.14. For a node v let $G(v)$ be the number of Gen nodes in its subtree and $\Pi(v) := \prod_u |t_u|$ over the $\forall E$ nodes u in its subtree, t_u the substituted term ($\Pi = 1$ if there are none). We show $|\text{concl}(v)| \leq (c_T \nu^2 + G(v)) \Pi(v)$ by induction on the tree. *Citation of a DT^o template* (of T , or A1–A3, A5, equality; substitution θ). An occurrence $M(\bar{t})$ becomes $\theta(M)[\bar{t}/\bar{z}]$. The body $\theta(M)$ has at most ν nodes and each argument t_i , a subterm of the template, has size at most a_T , so the piece has size at most νa_T . There are at most a_T occurrences and at most a_T rigid symbols, so $|\tau\theta| \leq a_T + a_T^2 \nu \leq c_T \nu \leq c_T \nu^2$. *Citation of A4* ($\forall x B \rightarrow B[t/x]$, not a template; ??). B and t have at most ν nodes each, so the size is at most $1 + (|B| + 1) + |B| |t| \leq 3\nu^2 \leq c_T \nu^2$ (such a tree has $\nu \geq 3$). *$\forall E$ with term t .* The conclusion $\psi[t/x]$ replaces each occurrence of the bound variable by t , which contains no bound indices, so no index shifting is needed: $|\psi[t/x]| \leq |\psi| |t| \leq |\text{premise}| |t|$, and Π gains the factor $|t|$. *Gen.* Replacing a parameter by an index keeps the size and the new $\forall$ adds one; G grows by one and $\Pi \geq 1$. *MP.* The conclusion is a proper subformula of the major premise, and G and Π only grow from a child to its parent. Finally $G(v) \leq \nu$, and the substituted terms of different $\forall E$ nodes are disjoint sets of grammar nodes, so $\sum_u |t_u| \leq \nu$. For k terms, AM–GM gives $\prod_u |t_u| \leq (\nu/k)^k \leq \sup_{y>0} (\nu/y)^y = e^{\nu/e}$. Hence $|\text{concl}(v)| \leq (c_T \nu^2 + \nu) e^{\nu/e} \leq (c_T + 1) \nu^2 e^{\nu/e}$. $\square$

Computed (c12_l1_size.out, Part B). Chains without A4 citations satisfy the stronger bound

$(c_T + 1)\nu e^{\nu/e}$ (the citation case gives $c_T\nu$). This is what the script tests: on 9865 conclusions of 3000 random valid $\forall E$ /Gen chains, with terms from a PCFG and cited sentences with up to three quantifiers and repeated variables, neither the inductive bound nor the final bound was violated; the largest $\ln(\text{size})/\nu$ was 0.220. The first version of the lemma had $\nu^{\nu+1}$ in place of $e^{\nu/e}$, and ν in place of ν^2 ; the A4 case (the referee's one-node example) forced ν^2 (model §12.4).

Proof of Proposition 1.15. The whole tree (derivation nodes, $\mathcal{Q}$ nodes, parameter-chain nodes) is a multi-type branching process with boundedly many children per node, each node's children drawn independently given its type. Give derivation nodes weight 1, $\mathcal{Q}$ nodes weight $\eta h_{\mathcal{Q}}$ ($h_{\mathcal{Q}}$ from ??) and chain nodes weight η . A derivation node has expected weighted offspring at most $m + \eta(v_T h_{\mathcal{Q}, \max} + 1)$, with v_T bounding the number of metavariables of a template of T or of K ; this is below 1 for small η , since $m < 1$. $\mathcal{Q}$ nodes satisfy the condition with $\rho_{\mathcal{Q}}$, and a chain node continues with probability $g < 1$. So the uniform condition holds with $\rho := \max(m + \eta(v_T h_{\mathcal{Q}, \max} + 1), \rho_{\mathcal{Q}}, g) < 1$, and ??(b),(d) gives $\mathbb{E}[e^{\zeta\nu}] \leq C$ for some $\zeta > 0$. A valid tree concluding φ has $\nu \geq \nu_{\min}(\varphi)$, so $P_T(\varphi) = \mu_T(\varphi)/Z_T \leq \Pr[\nu \geq \nu_{\min}(\varphi)]/Z_T \leq C e^{-\zeta\nu_{\min}(\varphi)}/Z_T$ by Markov's inequality. $\square$

Computed (c12_11_size.out, Part C). For the joint process with $m = 0.55$ and the grammar $\mathcal{Q}_A$ of the model appendix, 10^5 simulated trees: mean $\nu = 26.9$; $\Pr(\nu \geq L) = 0.362, 0.137, 0.0288, 0.00238, 3.0 \cdot 10^{-5}$ for $L = 25, 50, 100, 200, 400$; the log-tail falls by 0.0249 nats per node between $L = 100$ and 200.

Proof of Theorem 1.16. On input w , put $\ell := \ell(|w|)$ and guess a tree with at most ℓ grammar nodes. Each node is a choice from a finite list (rule, template of T , logical schema, production of $\mathcal{Q}$) or one of at most $\ell + O(1)$ variables, so the guess has $O(\ell \ln \ell)$ bits. Compute every node's conclusion explicitly, check validity (MP needs equality of formulas, $\forall E$ a root $\forall$) and compare the root's conclusion with φ_w . By Lemma 1.14 every formula has size at most $S := (c_T + 1)\ell^2 e^{\ell/e}$, and all operations are polynomial of a fixed degree d in that size. So the time is at most $\ell \cdot S^d$, which is at most $2^{(d+1)\ell}$ for $\ell \geq \ell_0(T)$, since $e^{d/e} < 2^d$ and the remaining factor is polynomial in ℓ ; smaller inputs go in a table. Completeness and soundness are as in Theorem 1.10: citations name a template of T , so no membership test is needed, and a valid tree is a K -derivation from $\bigcup \text{inst}(T)$ (each $\forall E$ is an A4 instance plus MP). $\square$

Proof of Corollary 1.17. Apply the hierarchy theorem with $t_1 := 2^{(d+1)s}$ and $t_2 := 2^{s^2}$; the condition $t_1(m+1) = o(t_2(m))$ is $s(m)^2 - (d+1)s(m+1) \rightarrow \infty$. This gives a decidable $X \in \text{NTIME}(2^{s^2}) \setminus \text{NTIME}(2^{(d+1)s})$. If, for some T and all but finitely many m , every $w \in X$ of length m had $\nu_{\min}(\varphi_w) \leq s(m)$, Theorem 1.16 with $\ell = s$ and a table (for the exceptional lengths and for $\ell < \ell_0(T)$) would put X in $\text{NTIME}(2^{(d+1)s})$. Since d does not depend on T , one X serves every T . The likelihood bound is Proposition 1.15 with $\nu_{\min}(\varphi_w) > s(m)$. $\square$

Proof of Proposition 1.18. (a) Summing over valid trees π concluding φ ,

$$\mu_T^\sigma(\varphi) = \sum_{\pi} \Pr(\pi) e^{-\kappa|\pi|} \leq e^{-\kappa\ell_{\min}^\sigma(\varphi)} \sum_{\pi} \Pr(\pi) = e^{-\kappa\ell_{\min}^\sigma(\varphi)} \mu_T(\varphi) \leq e^{-\kappa\ell_{\min}^\sigma(\varphi)} Z_T;$$

divide by Z_T^σ . (b) List the conclusions of all nodes of the tree in post-order (total size ℓ). For each $\forall E$ node v with premise u , insert the A4 instance $\text{concl}(u) \rightarrow \text{concl}(v)$ before v , of size $|\text{concl}(u)| + |\text{concl}(v)| + 1$; the substituted term has no bound variables, so it is free for x . Then $\text{concl}(v)$ follows by MP; Gen nodes are K 's Gen, and citations are axioms of K or members of $\bigcup \text{inst}(T)$. Each node is the premise of at most one node and there are at most ℓ nodes, so the insertions total at most

3ℓ and the derivation has size at most 4ℓ . If $\ell_{\min}^{\sigma}(\varphi_w) \leq s(m)/4$, there would be a K-derivation of size $\leq s(m)$; by Corollary 1.11(a) this fails for some $w \in X$ of length m , for infinitely many m . On those data (a) gives $-\ln P_T^{\sigma}(\varphi_w) \geq \kappa \ell_{\min}^{\sigma}(\varphi_w) - \ln(Z_T/Z_T^{\sigma}) > \kappa s(m)/4 - \ln(Z_T/Z_T^{\sigma})$. (c) An explicit K-derivation written with at least one bit per symbol has bit length at least its symbol size, which exceeds $s(m)$ on the same data. So $g(\ell_T(\varphi_w)) = 2^{-\kappa \ell_T(\varphi_w)} < 2^{-\kappa s(m)}$, and after normalisation $-\log_2 P_T^g(\varphi_w) = \kappa \ell_T(\varphi_w) + \log_2 Z_T > \kappa s(m) + \log_2 Z_T$. $\square$

Implicit in the notes. Part (c) needs the explicit derivations of ?? to be K-derivations (or derivations of a calculus that translates into K with linear overhead, which changes the constant) and the code to spend at least one bit per written symbol; the notes say “bits, which is at least their symbol count”. We state both assumptions.

The route to Conjecture 1.19, and what is missing. Represent conclusions with sharing (formula DAGs, grammar-compressed terms), so that $\forall E$ need not copy t into every occurrence; for the doubling family above a DAG of size $O(k)$ suffices. That every tree with ν grammar nodes has such representations of size polynomial in ν is part of what has to be shown. MP needs equality of two compressed formulas; equality of compressed strings and of grammar-compressed trees is decidable in polynomial time (recalled by the track, not checked: Plandowski 1994; Busatto, Lohrey and Maneth 2008; Schmidt-Schauß 2005). The missing step is Gen and de Bruijn indices under sharing: abstracting a parameter turns it into an index whose value depends on the binder depth at each occurrence, so a shared subterm can need different indices in different contexts. With that step, the proof of Theorem 1.16 would give $\text{NTIME}(\ell^c)$. Model §10 lists this as problem 12, together with the derivation-penalised equivalence (problem 13), a single-sorted repair of Hänni’s schema for consistent but unsound assigners (problem 14), and for which f the set A_f^C , or a set with its consequences and polynomial membership, is a finite union of DT^o templates (problem 15).

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